Design and Development of a Heart-Attack Detection Steering Wheel

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***Abstract*—Car accidents are becoming a growing concern to people with a history of chronic heart diseases. While some solutions have been proposed to detect the onset of heart attacks in automotive drivers, the existing applications are not comprehensive to provide the full accuracy of the detection process and encumber the driver to perform difficult tasks during a heart attack. Using pervasive technology, it is possible to collect in real-time the major symptoms at the onset of a heart attack. The heart rate, body temperature, and skin resistance can be quantitatively measured non-invasively; while, the qualitative symptoms of dizziness, nausea, fatigue, chest pain, and numbness in the left hand can be confirmed by the driver using an interactive system that non-intrusively asks the driver about his/her current physical state. In this sense, this paper describes the design, development, and validation of an instrumented steering wheel that guides an individual who is experiencing a heart attack through a safe process that may save his/her life and reduces the risks of accidents with bystanders or drivers in the vicinity. Data from various inputs are analyzed in real-time using a custom-developed smart algorithm implemented on an Arduino single-board microcontroller. If the algorithm determines that the user is at risk of a heart attack, the device performs the following actions automatically within 30 seconds: i) dispatches the current location of the driver to the paramedics with an SMS; ii) warns other drivers in the vicinity that there is a medical emergency situation on board; and, iii) asks the driver to park at the shoulder of the road and take a vasodilation pill from a built-in compartment. The developed system has undergone a rigorous bench-testing procedure and awaits to be tested in a real-life simulation. The originality of this device resides in the automation of the necessary tasks that a driver has to do in case of such emergency with minimum effort. Thus, the developed device provides an easy-to-use heart attack detection means that helps people with a history of chronic heart diseases to avoid car accidents while behind the steering wheel.**

Keywords – Car Accidents; Chronic Heart Diseases; Instrumented Steering Wheel; Myocardial Infarction; Road Safety.

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I. Introduction

Health-related car accidents are becoming a growing concern worldwide. Drivers with a history of chronic heart disease form a major category of prone individuals whose health ought to be continuously monitored since they are at risk of experiencing secondary attacks such as drowsiness, syncope, or myocardial infarction while driving. Detecting any unusual and abnormal health conditions and quickly assisting or intervening effectively is one of the key factors in reducing the incidence and severity of serious accidents attributed to health-related problems [1].

According to the American Heart Association (Dallas, TX, USA) and Mayo Clinic (Rochester, MN, USA), the recognized symptoms of a heart attack are: abnormal variations in temperature, blood pressure, and heart rate; sweat; nausea; fatigue; dizziness; chest pain; and, numbness in the left hand [2-3]. Hence, the continuous monitoring of these symptoms forms the foundation of a smart system that automatically detects the onset of myocardial infarction in automotive drivers.

The aim of this study was to design and develop an interactive steering wheel system that detects the signs of an approaching heart attack in individuals with a history of chronic heart diseases while driving. The developed device continuously monitors the driver's heart rate, body temperature, and skin resistance noninvasively. While, the qualitative symptoms of dizziness, nausea, fatigue, chest pain, and numbness in the left hand are confirmed by the driver using an interactive system that un-intrusively asks the driver about his/her current physical state. Once the onset of a myocardial infarction is detected, the driver is guided, through a smart algorithm, to park the vehicle on the shoulder of the road, while the device performs the following actions automatically within 30 seconds: i) dispatches the

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current location of the driver to the paramedics with an SMS; ii) warns other drivers in the vicinity that there is a medical emergency situation on board; and, iii) dispenses to the driver a vasodilation pill from a built-in compartment. The rationale behind such a process is to save the driver's life while also providing road safety.

An additional originality of this device is rendered through the automation of the necessary tasks that a driver has to do in case of such emergency with minimum effort. Thus, the developed device provides an easy-to-use heart attack detection means that helps people with a history of chronic heart diseases to prevent car accidents while behind the steering wheel.

II. Literature Review

Leijdekkers and Gay have developed a heart attack self-test application for a mobile phone that allows potential heart attack victims to rapidly assess whether they are experiencing a heart attack, without the need of medical specialists' intervention [4]. The application uses pervasive mobile technologies in conjunction with ECG monitoring to detect the onset of a heart attack. Thereafter, the application urges the victim to call the emergency service so as to automatically dispatch the current location of the victim.

Futatsuyama *et al.*, designed a built-in vital sign sensor that permits drivers to non-invasively measure their heart rate and pulse waveform using a pulse sensor mounted on the steering wheel of a motor vehicle. Despite the use of a non-gelled sensor and the noisy environment of the motor vehicle, the authors were able to suppress the mechanical vibration as well as the electrical noise affecting the signal obtained from the pulse wave sensor through various design techniques such as anti-vibration mounting mechanisms and ensemble averaging of the acquired signal [1].

Vavrinsky *et al.* presented the design of a health monitoring system for car drivers using thin film micro-sensor with Inter Digital Array Transducer (IDAT) technology [5]. The newly developed multipurpose IDAT micro-sensor consists of an impedance sensor integrated with a temperature resistive sensor mounted on a single chip. This multipurpose sensor permits the monitoring of psychogalvanic reflex, heart pulse, and skin temperature of a car driver, using depth impedance analysis of different skin layers by selecting the appropriate size of microelectrodes [5].

Ashrafuzzaman *et al.* proposed a system that is capable of measuring the heart rate, using a smart phone camera and a mobile stethoscope to record heart sound for the detection of a heart attack as well as some other heart-related disorders [6]. The detection process is implemented via a Fuzzy Logic algorithm. The system requires that the user places his/her index finger on the built-in camera of a smart mobile phone so as to derive the heart rate; it also requires the user to place the stethoscope over the chest area so as to determine the various Korotkoff heart sounds and other anomalies such as heart murmurs.

In 2012, Texas Instruments presented a proof of concept as to how biometric sensors mounted on a steering wheel may be used to obtain vital information from a driver in real time [7]. This design combines modern solid-state technology with low-power processing ability and wireless communication to measure pulse rate, respiration rate, and ECG-based heart rate from a standalone system. The three key biometric measurements are then evaluated by a remote TI Health HUB application running on an iPad.

In 2014, Patel and Chauhan reported on the utilization of a commercially available gaming device that can perform the dual function of detecting a heart attack and notifying the emergency services. The article first illustrates how a Kinect motion sensing input device—developed by Microsoft Corporation, Redmond, Washington, United States—can be incorporated into the detection process of a heart attack. Then it describes how features such as a GSM system and Video Conferencing could be used in sending an alert SMS to—and then communicating with—relatives and emergency services [8].

Yadav and Gowda, in 2016, developed a wearable device that is intended for the early detection of a heart attack. The device makes use of commercially available smart health bands along with a smart mobile phone to continuously monitor the heart rate of the user. Should the heart rate drop below a pre-set threshold value, an application installed on the mobile phone will transmit an alert message to both the ambulance services and other specified contacts [9].

In 2017, Leem *et al.* reported on the development of a vital-sign monitoring mobile system used in conjunction with Impulse Radio Ultra-Wide Band (IR-UWB) radar [10]. The paper proposes a new algorithm that can extract the vital signs from the motion-artifact contaminated signals acquired non-invasively during the driving activities. It also reduces the possibilities of car crashes by utilizing novel techniques via the (IR-UWB) radar to detect drowsiness and attempts of mobile phone usage while driving.

In 2017, Sidheeque *et al.* reported on the implementation of a system that monitors the heart rate and detects heart attacks [11]. The system employs a pulse-sensing unit interfaced to a microcontroller that permits the sampling of the heart rate signals and transmitting them over the Internet to a remote console for live monitoring and emergency response.

Even though the above studies present state-of-the-art work, they do not cover the full spectrum of aforementioned symptoms in the assessment of myocardial infarctions [2-3]. Additionally, certain studies may encumber the user with various means while acquiring the sought data. Accordingly, the present study has taken these undesired issues into consideration while optimizing the design requirements.

III. Materials and Methods

The system, designed and developed in the current study, consists of a standard steering wheel instrumented with various sensors: i) a heart rate sensor that uses infrared radiation to detect the heart rate of the driver; ii) a

temperature sensor that works by focusing the infrared energy emitted by the driver onto a photodetector; iii) a skin impedance sensor that detects the amount of perspiration of the driver; and, iv) a touch sensor that is used to activate all aforementioned sensors. An Arduino Mega 2560 microcontroller is used with a custom developed smart algorithm to analyze the data streamed from the various sensors, and to control the following connected peripherals: i) a TFT screen mounted on the dashboard to display the data measured by the various sensors, and a shield to interface the screen with the Arduino board; ii) a GPS module to determine the exact location of the driver's vehicle; iii) a GSM shield to connect the system to a cellular communication network; iv) a relay module connected to a 30 cm x 20 cm LED screen to warn other drivers in the vicinity that there is a medical emergency situation on board; v) a servo motor to control the door opening of the medication dispenser compartment; and, vi) an audio system that includes a speaker, an MP3 player, a sound amplification circuit, and an SD card to provide high capacity memory for the storage of pre-recorded audio files. The entire low-power consumption system is powered by the 12-Volt car battery. Fig. 1 shows the block diagram of the developed system.

The integrated system starts by measuring the heart rate, body temperature, and skin impedance. These inputs are taken from two different locations on the steering wheel. The data from the sensors is multiplexed at a sampling rate of 25 samples per second (sps) for each sensor, and are compared to pre-set threshold values based on the following norms:

- According to the American Heart Association, the maximum average heart rate is 100 bpm [12].
- According to the University of Rochester Medical Center (URMC), the minimum average body temperature is 37 °C [13].
- According to the U.S. National Institute for Occupational Safety and Health (NIOSH), under dry conditions, the resistance offered by the human body may be as high as 100 kΩ. Wet or broken skin may drop the body's resistance to 1 kΩ. Therefore, the average of the minimum and maximum skin resistance is 50.5 kΩ [14].

Table I provides comprehensive information relevant to the symptomatic signs of a heart attack; and, hence, it has been adopted in this study [2, 3, 12-14].

Subsequently, and by the following order of priority, if the heart rate is found to be more than the average maximum, the system stores this value and proceeds to measure body temperature. If the latter is found to be below the threshold value, the system stores the value and proceeds to measure the skin impedance. If this is found to be less than the average, the system stores this value and moves on to acquire the qualitative data, whereby, the system verbally asks the driver the following five questions related to his/her current physical state: i) the presence of chest pain, ii) feeling of numbness in the left hand, iii) feeling of nausea, iv) feeling of dizziness, and v) feeling of fatigue.

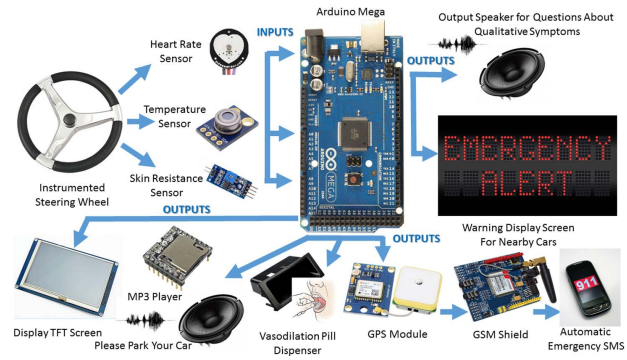

Fig. 1. A block diagram of the system's design.

Table I. Information relevant to the symptomatic signs of a heart attack.

▪ **Fluctuations in Heart Rate:** Having a heart rate faster or slower than normal of more than 100 beats per minute (BPM) or less than 60 BPM is one of the most recognized signs that lead to a heart attack [12].
▪ **Fluctuations in Body Temperature:** Abrupt temperature changes are common and recognized in heart attacks [13].
▪ **Cold Sweat:** Sweating or perspiration can accompany a heart attack where it is described by its cold sweats in the upper body [14].
▪ **Nausea:** Nausea or feeling sick in the stomach as well as belching or burping can accompany the nausea. Also, indigestion is associated with a heart attack [2-3].
▪ **Fatigue:** Feeling fatigued or winded after doing something that there was no problem doing it in the past could be a sign of heart attack [2-3].
▪ **Dizziness:** Feeling dizzy or lightheaded suddenly means that the blood pressure has dropped down. This sign can accompany a heart attack [2-3].
▪ **Chest Pain:** Chest pain or Angina involves discomfort in the center of the chest that lasts more than a few minutes, or that goes away and comes back. It may feel like uncomfortable pressure, squeezing, fullness or pain [2-3].
▪ **Numbness in the Left Hand:** The pain of a heart attack can spread down to the arms—Referred Pain—especially in the left arm where numbness occurs [2-3].
▪ **Very High/Low Blood Pressure:** Sometimes, blood pressure can decrease before a heart attack which can be due to the response to pain before the heart attack. Also, parasympathetic nervous system which is responsible for the body's resting state goes into overdrive. High blood pressure is also accompanied to a heart attack that is caused by spikes in hormones like adrenaline that flood the body during stressful situations like heart attacks [2-3].

The captured data is then analyzed in real-time using a smart algorithm specifically designed to accurately assert the presence of a heart attack. Once the algorithm identifies a positive detection, it performs the following actions: i) signals the system to send a distress signal via SMS to the paramedics with a transmission of the vehicle's geographical location; ii) displays an informative warning message for nearby drivers to ensure road safety; iii) asks the driver to immediately park the vehicle on the shoulder of the road; and, iv) asks the driver to take a vasodilation pill dispensed from a built-in compartment on the dashboard. A flowchart showing the algorithmic details of the developed system is illustrated in Fig. 2.

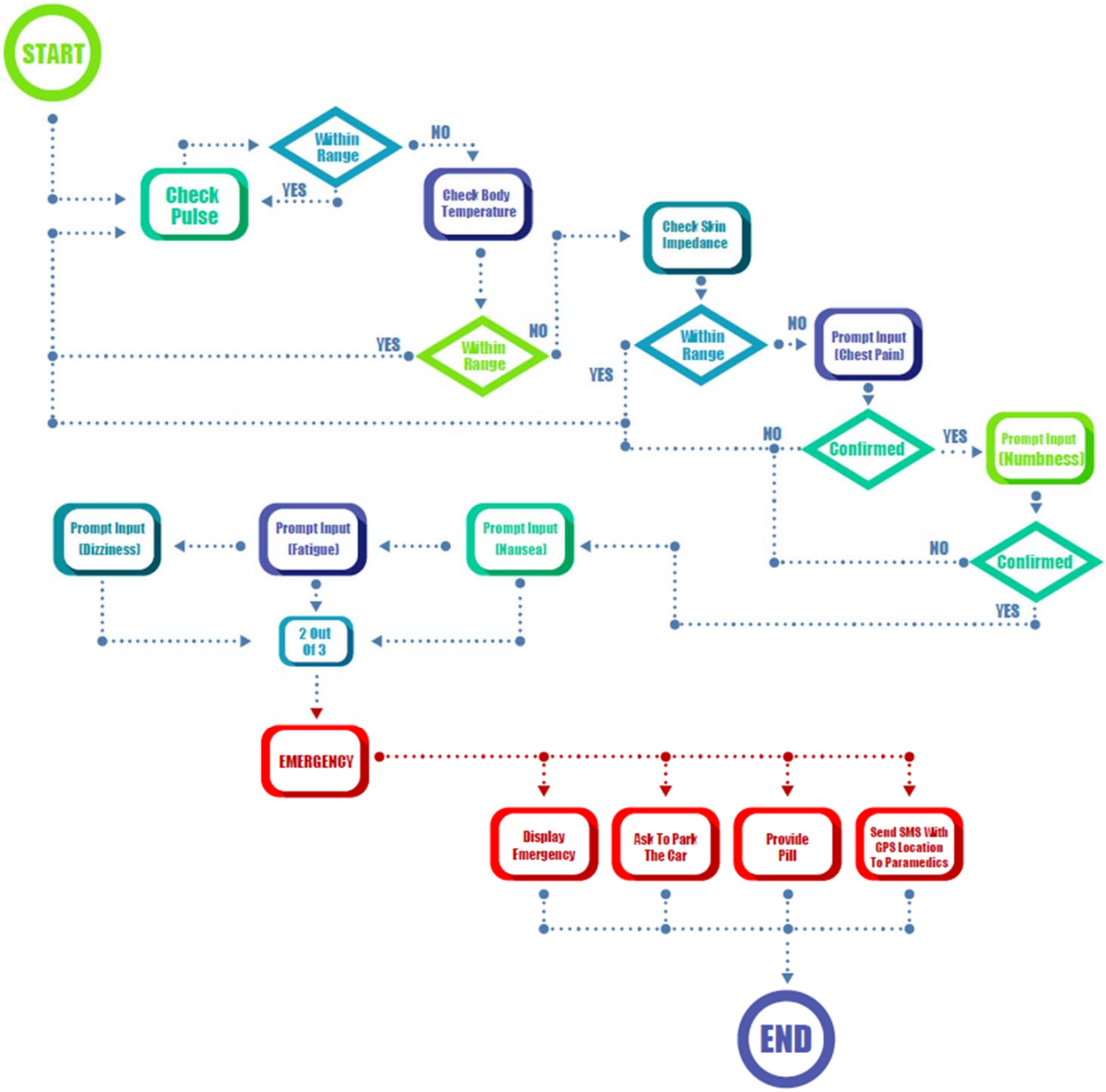

Fig. 2. A flowchart showing the algorithmic details of the design.

IV. Results

The developed system underwent rigorous controlled bench testing procedures. Each constituent sensor of the instrumented steering wheel was tested in the normal and abnormal ranges of its corresponding physiological measurand. Different manipulation techniques were used on each sensor alone as well as in combination with the others to produce a situation that simulates a heart attack, following the logic presented in Fig. 2. These manipulations were above and below the aforementioned average value specific for each symptom. Subsequently, the qualitative inputs were manipulated to yield a heart attack setting, in synch with the quantitative inputs. Additionally, all system outputs were randomly tested and their various possible outcomes were confirmed. These testing sequences asserted that the system designed and developed in this study can detect—with a maximum of 30 seconds—the symptoms of a myocardial infarction before it occurs. Table II illustrates a typical scenario of the system's response to an individual experiencing a heart attack while driving. Certainly, other plausible scenarios also exist and are accounted for by the system.

V. Discussion and Conclusions

An instrumented steering wheel system designed and developed for the detection of heart-attacks in drivers with a history of chronic heart diseases has been validated. The system identifies eight out of the nine recognized symptoms of a heart attack [2-3]. The measurement of blood pressure was excluded from this protocol because of the lack of availability of a non-intrusive commercially-available sensor that measures this vital parameter.

Current methods used in the indirect measurements of blood pressure are based on variants of the auscultatory technique [15]. This technique employs an occlusive cuff—surrounding the brachial or radial artery—that is inflated until the pressure is above systolic pressure, and then is slowly deflated at a rate of 2-3 mm Hg. Systolic pressure is determined at the onset when the blood spurts beneath the cuff and causes an audible pulse that is detected via a stethoscope. Diastolic pressure is then determined when the last audible pulse is detected. Although, automated blood pressure measuring devices that are based on the auscultatory technique are commercially available and offer the means to assess blood pressure, they are considered to be encumbering to the driver. Hence, variations in blood pressure were not measured in this study and remain a target for future development by our research team. Even though this shortcoming slightly affects the accuracy of the system, the eight measured parameters provide a superior edge to any developed system reported in the surveyed literature [1, 4-11]; hence, it is deemed to be sufficient at present to accurately determine the onset of a myocardial infarction.

Future developments will also include further automated testing of the system, and the use of Fuzzy Logic in the detection algorithm in place of the current one that is based on Crisp Logic. The developed system has been designed with the intention of saving the lives of individuals with a history of chronic heart diseases in the event of a heart attack during driving, while also ensuring road safety to other drivers happening to be in the vicinity of the sufferer's vehicle.

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Table II. A typical scenario of the system's response to an individual experiencing a heart attack while driving.

System Quantitative Inputs:
- Heart Rate measurement: 110 bpm (High)
- Body Temperature measurement: 36.0 °C (Low)
- Skin Resistance measurement: 8000 Ω (Low)

System Qualitative Inputs:
- Chest Pain? ***Confirmed by Driver***
- Numbness in Left Hand? ***Confirmed by Driver***
- Nausea? ***Confirmed by Driver***
- Fatigue? ***Confirmed by Driver***
- Dizziness? ***Confirmed by Driver***

System Outputs:
- Back LED Display:
  Caution! Medical Emergency on Board
- Voice Message:
  Please park your car on the shoulder of the road
- Automatic Emergency Message:
  Please wait in the car! A message has been sent to Paramedics with the geographical location of the car
- Voice Message:
  A vasodilation pill has been dispensed for you. Please, take it and put it beneath your tongue!

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